

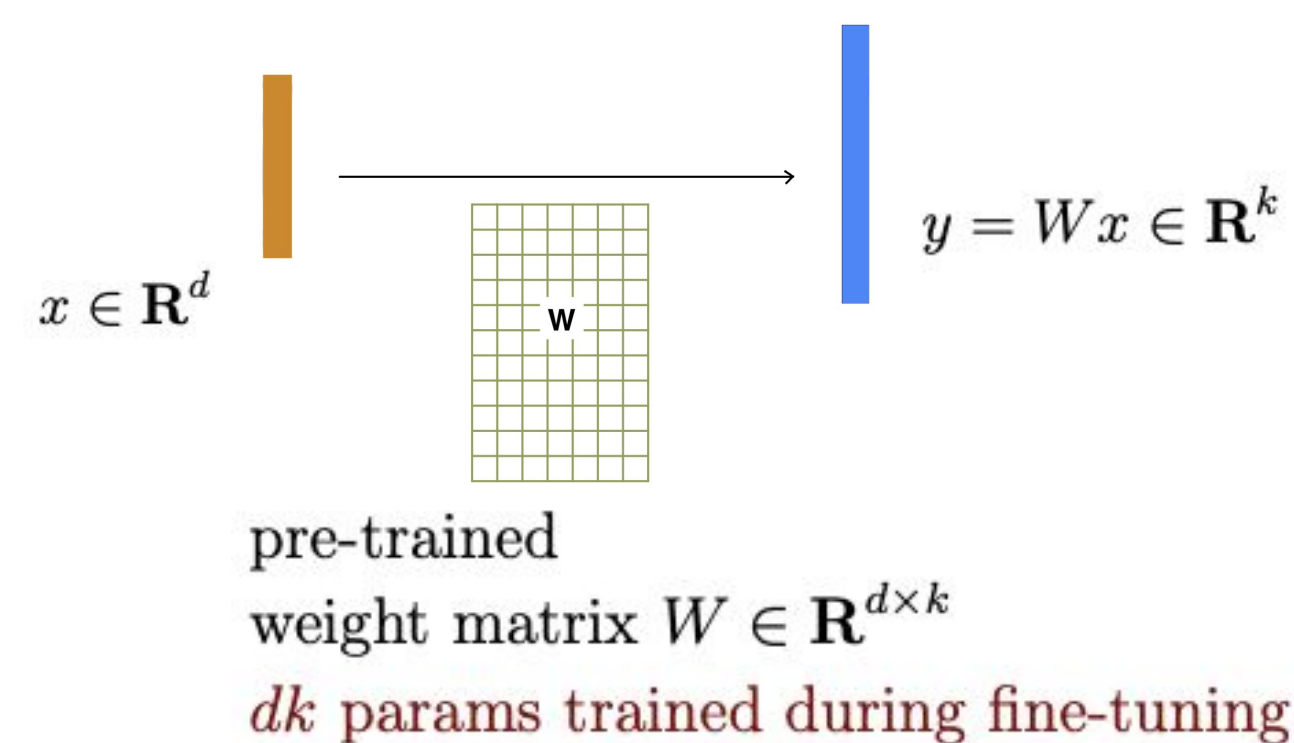
Reimplementation of LoRA: Low-Rank Adaptation of LLMs

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Introduction

• Problem

Adapt a pre-trained LLM to multiple tasks by fine-tuning.
Full fine-tuning is too expensive since it updates ALL the parameters.



• Context and motivation

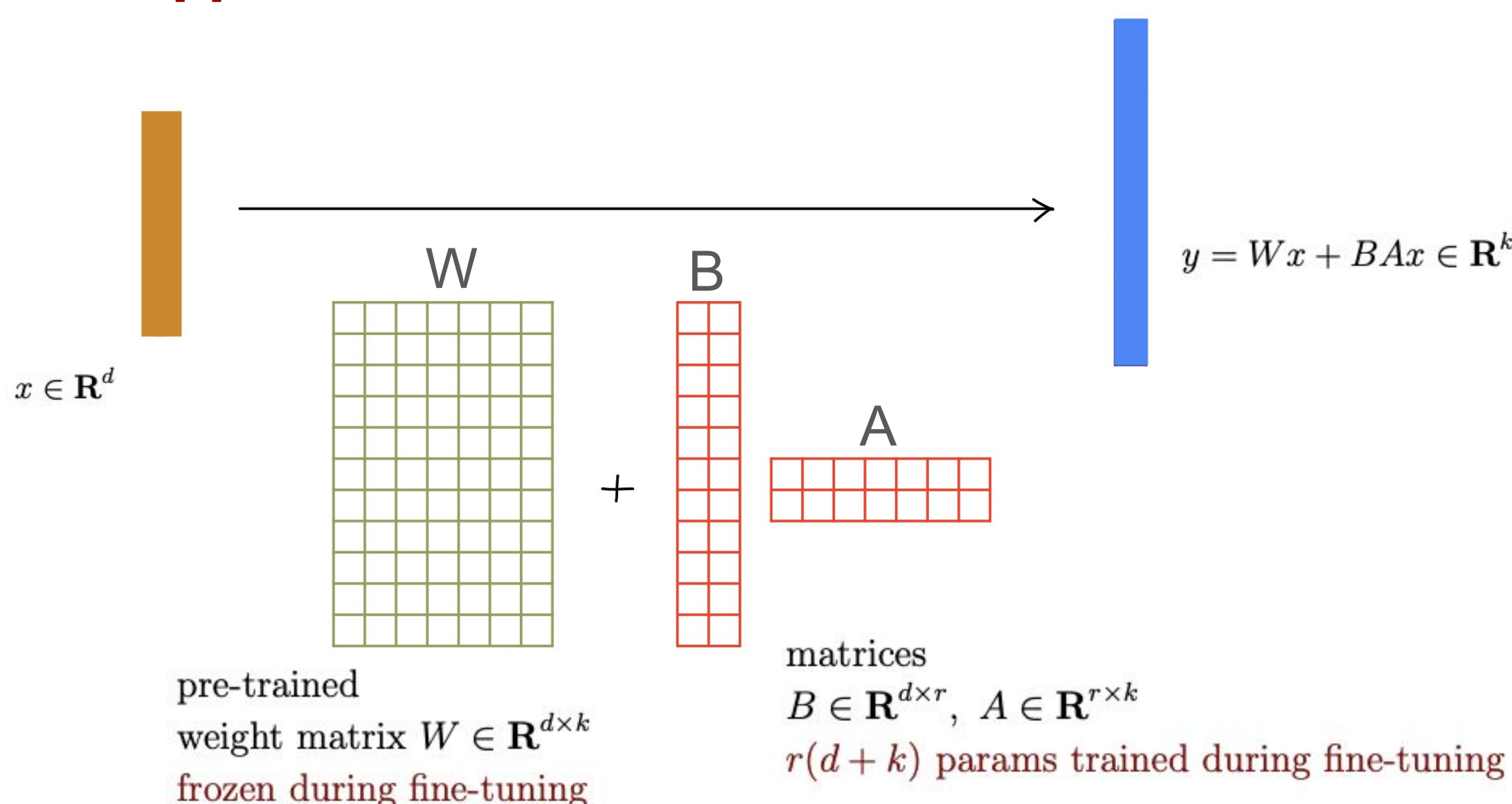
LLMs are too large: RoBERTa-Large has 355M params, GPT-3 has 175B. It is not feasible to store multiple copies of the model to full fine-tune for separate tasks.

• Goal

Reduce the number of parameters used in fine-tuning, while keeping high accuracy

• Summary of LoRA [1]

LoRA is a parameter-efficient fine-tuning method for LLMs. Instead of learning dk params, we learn $r(d+k)$ params. In [1], rank $r=8$.



• Methodology

Replace the layers in RoBERTa-Large with LoRA layers

Models: RoBERTa-Large

Datasets: GLUE

Tools used: Adam optimizer, RoBERTa-Large, GLUE dataset, code in [2] for inspiration

• Challenges encountered during re-implementation

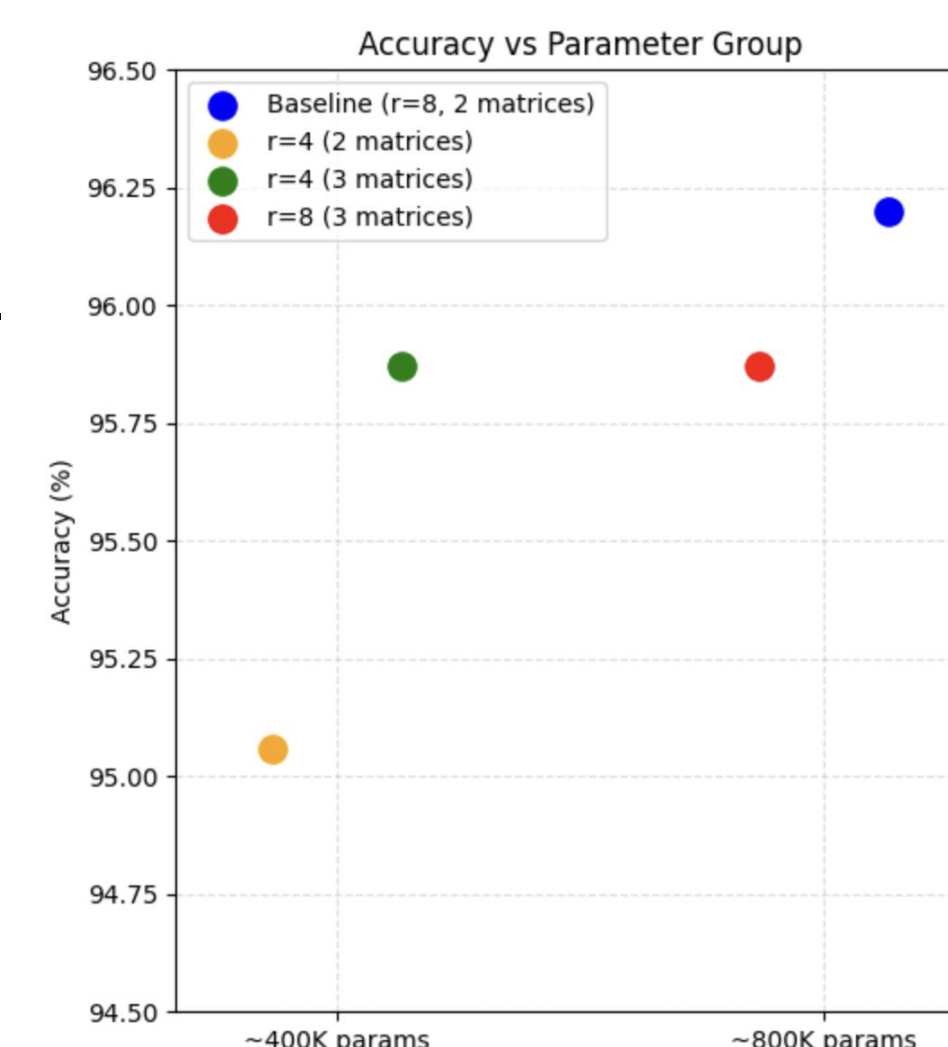
Converting the RoBERTa-Large layers to LoRA layers

Results & Benchmarks

- For RoBERTa-Large with 355M params, reduce the number of params used in fine-tuning to about 0.8M, while keeping 96% validation accuracy.

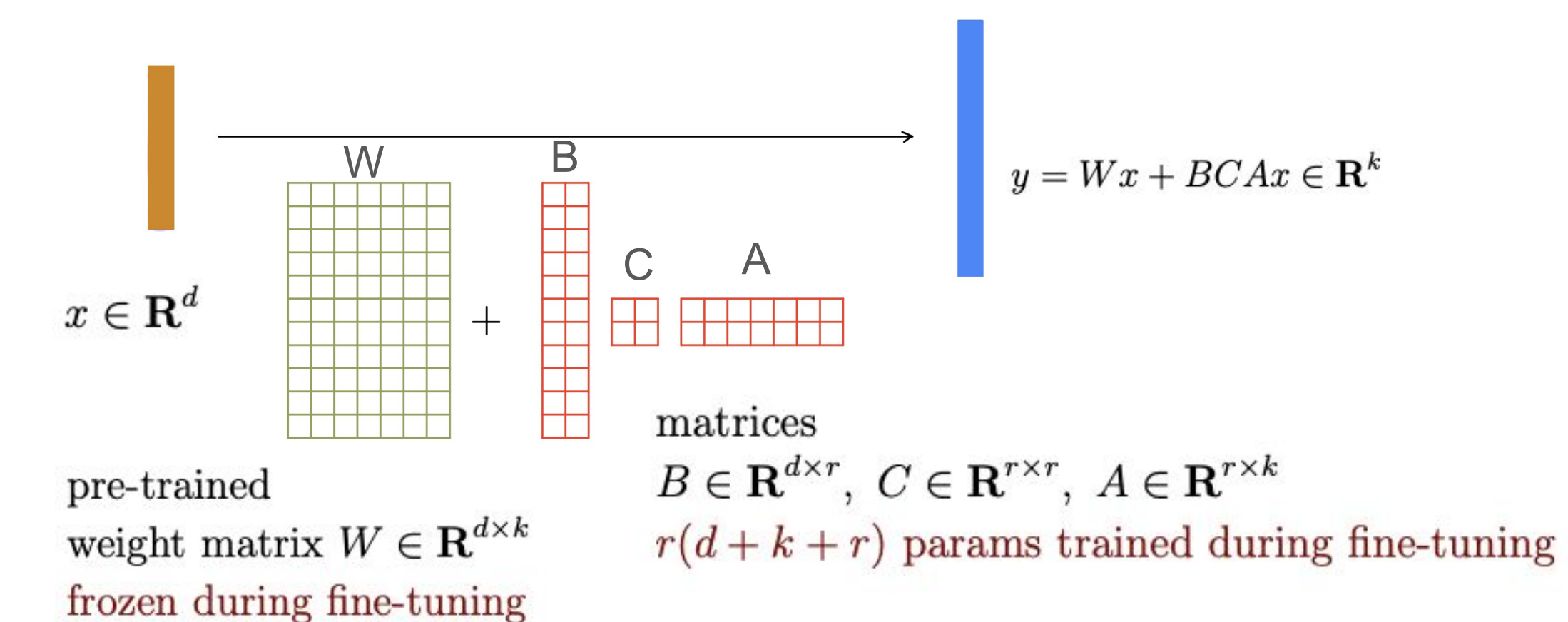
	type of fine-tuning	trainable params in fine-tuning	percentage of trainable params	validation accuracy
RoBERTa-large	full fine-tuning	355,755,010	100%	96.4%
results in [1]	LoRA rank 8	800,000	0.22%	96.2%
My results:				
[1] reimplemented	LoRA rank 4	395,266	0.11%	95.06%
with additional third matrix	LoRA rank 4	396,034	0.11%	95.87%
with additional third matrix	LoRA rank 8	791,554	0.22%	95.87%

Table1: Pre-trained model with 355,755,010 params is fine-tuned. Table shows the trainable params and validation accuracy of the models.



• Analysis of results in the context of [1]

- Explored rank $r=4$ instead of $r=8$ in [1]. Due to lower rank, accuracy 95.06% is lower, but the advantage is half of the params used.
- Explored adding an extra (third) matrix and rank $r=4$. It still uses half the params used for $r=8$, but improves accuracy to 95.87%, very close to 96.2% in [1].



Conclusion

LoRA can drastically reduce the number of parameters in fine-tuning while keeping high accuracy, allowing the model to be fine-tuned with much lower storage and compute costs.

Directions for Future Work

- Understand better why LoRA works
- Find methods how to pick the matrices (to apply LoRA to), instead of using heuristics



References

- [1] LoRA: Low-Rank Adaptation of Large Language Models, Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, Weizhu Chen, *International Conference on Learning Representations (ICLR) in 2022*, arXiv: 2106.09685.
- [2] Code Repository on GitHub: github.com/microsoft/LoRA
- [3] ChatGPT was used to produce the cartoon.